

UNIVERSIDADE FEDERAL DO ABC / UFABC
Introduction to Economics

CHOCOLATE:
Value Addition in the Cocoa Chain and the Opportunity
of the Japanese Market for Brazil

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Abstract

This paper examines Brazil's position in the global cocoa and chocolate value chain, discussing why exporting the industrialized product tends to capture more value than exporting the raw material. Using official foreign-trade data, the study compares Brazilian exports of cocoa and chocolate, their main destinations, and the size of their respective world markets. It then examines the Japanese chocolate market — one of the largest global importers, where Brazil's share is still marginal — and discusses how the reciprocal visa exemption between Brazil and Japan, in effect since September 2023, may reduce transaction costs and favor closer trade ties between the two countries. The paper concludes that Brazil holds a comparative advantage in cocoa production but has not yet fully converted it into a competitive advantage in chocolate exports, which represents both a challenge and a trade-policy opportunity.

Keywords: international trade; value chain; cocoa; chocolate; Brazil; Japan.

1. Introduction

Chocolate is a high value-added product whose production chain leaves room for large, medium, and small producers alike, generating direct and indirect jobs across its entire span — from cocoa cultivation to industrial processing and distribution of the final product. These features make the sector an interesting case for the economic analysis of international trade, since they allow one to observe, within a single production chain, different degrees of value addition and different ways in which the same country can be inserted into the world market.

Brazil holds a prominent position as a producer and exporter of cocoa and its derivatives, but — as this paper aims to show — it still exports the product mainly at less processed stages of the chain. At the same time, a recent development that has received little attention in the economic literature on the subject — the reciprocal visa exemption between Brazil and Japan, in effect since September 30, 2023 — opens a window of opportunity for closer trade ties between the two countries, including in the chocolate market, where Brazil's presence is today practically negligible.

The objective of this paper is (i) to situate Brazil within the global cocoa and chocolate value chain, comparing exports of the raw material with exports of the industrialized product; and (ii) to assess the potential of the Japanese chocolate market as a destination for Brazilian exports, in light of the recent reduction in mobility barriers between the two countries.

2. Theoretical Framework

The theory of comparative advantage, formulated by David Ricardo, explains why countries specialize in producing and exporting the goods in which they have the lowest relative opportunity cost. Brazil's

agroclimatic conditions — especially in the Amazon region and southern Bahia — have historically favored cocoa cultivation, which explains the country's position among the world's leading producers of the commodity.

However, holding a comparative advantage in producing a raw material does not, by itself, guarantee the capture of the largest share of the value generated along the production chain. The concept of the value chain, developed by Michael Porter, shows that each link of a production process — from raw-material extraction to processing, distribution, and branding — adds a distinct amount of economic value. In agro-industrial chains such as cocoa's, it is widely observed that the industrial-transformation stage (processing cocoa into chocolate, with branding, packaging, and quality standardization) tends to retain higher margins than the agricultural stage.

Add to this the role of transaction costs in international trade: non-tariff barriers, such as visa requirements for business travelers, raise the cost of prospecting new markets, of trade missions, and of building long-term business relationships. The international-economics literature (for example, Krugman, Obstfeld, and Melitz) recognizes that reductions in this type of barrier, even when non-tariff in nature, tend to facilitate bilateral trade by lowering the costs of market access.

3. Brazil in the Global Cocoa and Chocolate Chain

3.1 Brazilian production and standing

According to data from ApexBrasil (the Brazilian Trade and Investment Promotion Agency), Brazil is the world's seventh-largest producer of cocoa and its derivatives and the seventh-largest exporter of these products, serving more than 135 countries. Brazil is one of the few countries in the world that holds the entire chocolate chain vertically integrated at home — from cocoa cultivation to the production of high-quality bars — a structural advantage that is uncommon among major agricultural-commodity producers. Between 2021 and 2022, Brazilian exports in the sector grew 12.7%, signaling an expanding trajectory in foreign trade.

3.2 Cocoa vs. chocolate: comparing value addition

The table and charts below compare Brazilian exports of cocoa (at raw or semi-processed stages — raw beans, powder, roasted beans, butter, and paste) with exports of chocolate (the final industrialized product), for the year 2022.

Indicator	Cocoa (raw, powder, roasted, butter, paste)	Chocolate (final product)
Brazilian exports (2022)	US\$ 228 million	US\$ 142 million

Main destinations	Argentina (51%), Chile (23%), USA (20%)	Argentina (27%), Chile (11%), Paraguay (10.7%)
World market	US\$ 23.1 billion	US\$ 33.3 billion

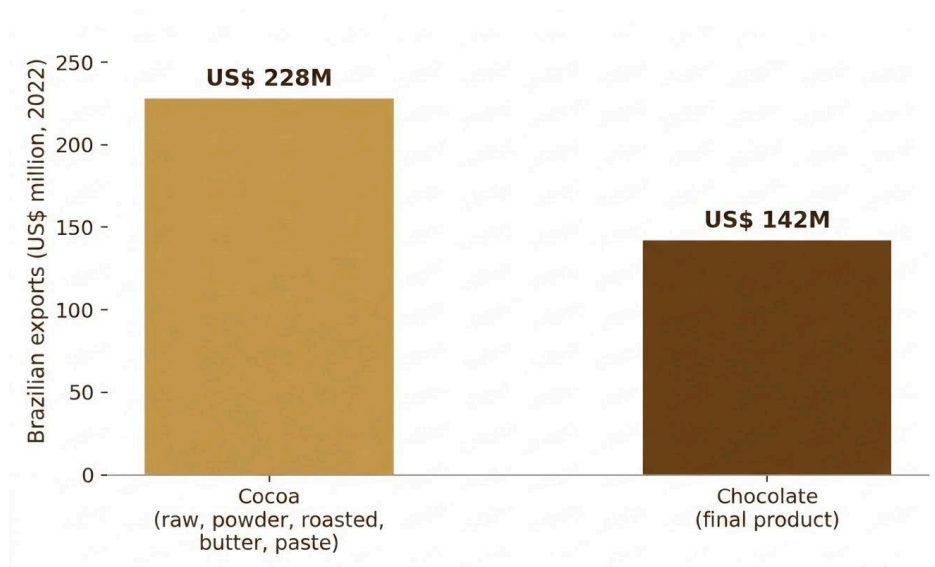


Chart 1 — Brazilian exports of cocoa and chocolate, 2022 (US\$ million). Source: prepared by the author based on ApexBrasil data (2023).

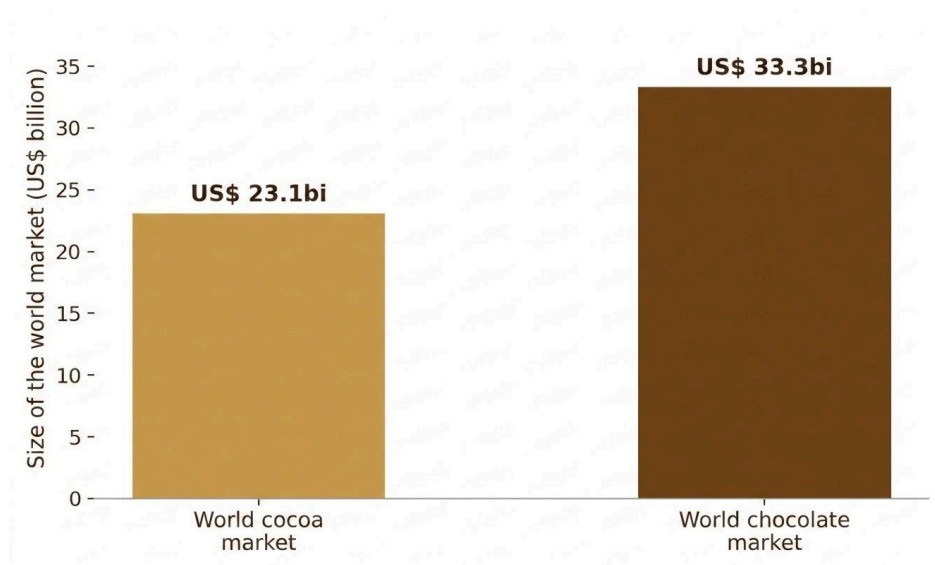


Chart 2 — Size of the world cocoa and chocolate markets (US\$ billion). Source: prepared by the author based on ApexBrasil data (2023).

A first point worth noting is that, even though Brazil exports more raw or semi-processed cocoa (US\$ 228 million) than chocolate itself (US\$ 142 million), the world chocolate market (US\$ 33.3 billion) is

significantly larger than the world cocoa market (US\$ 23.1 billion). This pattern reflects the international market's preference for the manufactured product and suggests that Brazil concentrates its exports precisely in the link of the chain where the world market is relatively smaller and, presumably, of lower value added per unit exported.

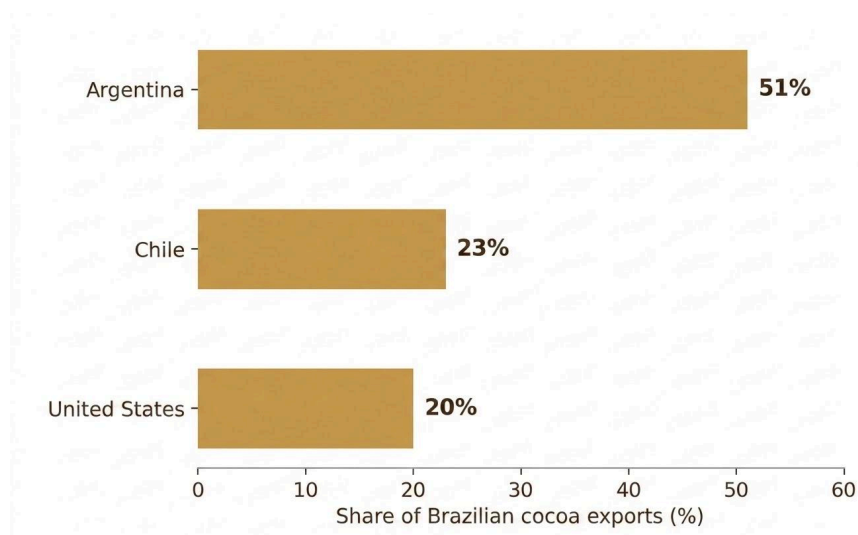


Chart 3 — Main destinations of Brazilian cocoa exports (%). Source: ApexBrasil (2023).

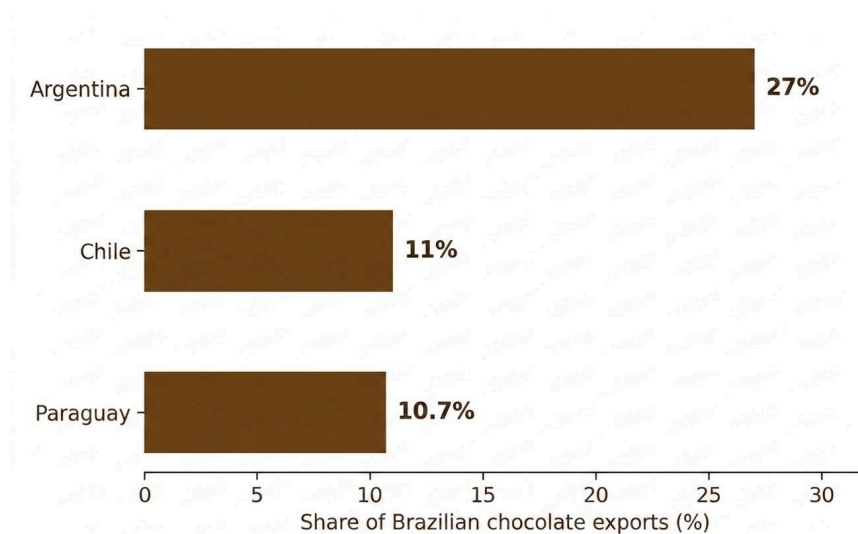


Chart 4 — Main destinations of Brazilian chocolate exports (%). Source: ApexBrasil (2023).

The destinations of Brazilian exports, for both cocoa and chocolate, are heavily concentrated in neighboring South American countries — Argentina, Chile, and Paraguay — with the United States appearing as a relevant destination only for raw cocoa. This geographic concentration points to still-limited market diversification, especially for the higher value-added product, and reinforces the relevance of examining more geographically distant, yet large, markets such as Japan's.

4. The Japanese Market and Brazil-Japan Relations

4.1 Profile of the Japanese import market

According to data from The Atlas of Economic Complexity (Growth Lab, Harvard Kennedy School), Japan imports approximately US\$ 587 million worth of chocolate per year. Of that total, only about US\$ 46 million (roughly 7.8%) originates from the Americas as a whole, and Brazil's share of this market is just US\$ 1.11 million — the equivalent of 0.19% of Japan's total imports.

Origin of Japanese chocolate imports	Value (US\$)	Share
Total imported by Japan (world)	587 million	100%
Origin: the Americas	46 million	≈ 7.8%
Origin: Brazil	1.11 million	0.19%

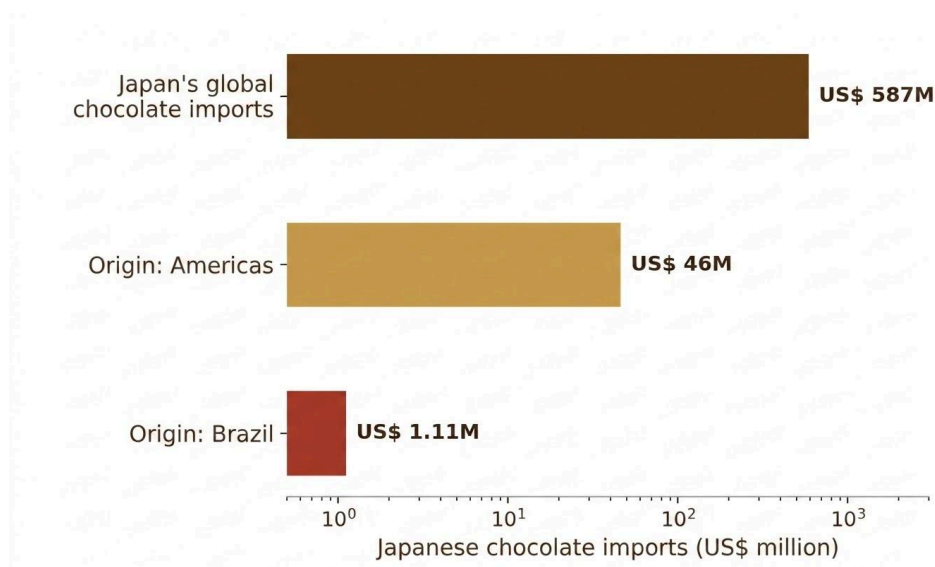


Chart 5 — Japanese chocolate imports: from the world total to Brazil's share (US\$ million, log scale). Source: prepared by the author based on *The Atlas of Economic Complexity*.

These figures indicate that the Japanese chocolate market is dominated by traditional suppliers — mostly European, such as Belgium, Italy, France, Germany, and Switzerland — and that Brazil's presence today is residual, even though Brazil is a significant cocoa producer on a global scale. This is, therefore, a very large market with extremely low Brazilian penetration, which can be read both as a sign of difficult access and as a still-untapped space for growth.

4.2 The Brazil-Japan visa exemption (2023) as a driver of closer trade ties

In August 2023, the Japanese government confirmed its decision to exempt Brazilian citizens from visa requirements for short-term tourism and business travel (up to 90 days), effective September 30, 2023. In reciprocity, Brazil stopped requiring visas from Japanese citizens. Both governments framed the measure as part of strengthening the relationship between the two countries as "global strategic partners," within the context of multilateral agendas such as the G20.

From an economic standpoint, lowering mobility barriers tends to reduce the transaction costs associated with trade prospecting: business trips, trade missions, participation in industry fairs, and in-person negotiations become simpler and less costly. For a product like chocolate — whose sale in the premium retail segment usually depends on direct business relationships, tastings, certifications, and brand-building — this kind of mobility facilitation can be particularly relevant, even though it does not, by itself, replace the structural challenges of market access.

5. Discussion: Opportunities and Challenges for Brazil

Taken together, the data presented make it possible to identify opportunities and challenges for a potential Brazilian strategy to expand into the Japanese chocolate market:

- Opportunity — room to grow: Brazil's share of just 0.19% of the Japanese chocolate market, combined with the fact that Brazil already holds the entire production chain from cocoa to chocolate, suggests growth potential from a very low base.
- Opportunity — lower transaction costs: the reciprocal visa exemption, in effect since September 2023, makes it easier to conduct trade-prospecting trips, business missions, and in-person negotiations between the two countries.
- Opportunity — differentiation through origin and sustainability: attributes such as Amazonian origin, agroforestry practices, and traceability can serve as a competitive differentiator against traditional suppliers.
- Challenge — competition from established brands: the Japanese market is heavily dominated by well-established European chocolates (Belgium, Italy, France, Switzerland, Germany), with strong brand recognition among consumers.
- Challenge — quality standards and certification: access to Japanese retail usually requires specific sanitary and quality certifications, as well as packaging and labeling adapted to the local market.
- Challenge — logistics and scale: the geographic distance and the logistics costs associated with transporting a temperature-sensitive product like chocolate impose additional price-competitiveness challenges.

These factors suggest that lowering mobility barriers is a favorable condition, but not a sufficient one: it needs to be accompanied by a deliberate trade strategy — for example, through export-promotion programs such as those run by ApexBrasil — for the opportunity opened by the visa exemption to actually translate into higher Brazilian chocolate exports to Japan.

6. Concluding Remarks

This paper has shown that Brazil holds a comparative advantage in cocoa production, underpinned by favorable agroclimatic conditions and by the fact that it holds — unusually among major agricultural-commodity producers — the entire chocolate production chain. Nevertheless, the country still exports proportionally more raw material than industrialized product, forgoing part of the value added that the international market attributes to chocolate — whose world market is roughly 44% larger than that of raw cocoa and its derivatives.

In the specific case of the Japanese market, one of the world's largest chocolate importers, Brazil's presence today is practically nonexistent. The reciprocal visa exemption between Brazil and Japan, in effect since September 2023, represents a favorable factor for closer trade ties between the two countries by reducing transaction costs linked to business mobility. Even so, effectively winning this market depends on additional factors — certification, adaptation to local taste and standards, brand-building, and logistics — that fall outside the scope of this paper but constitute a relevant agenda for future research in international economics and foreign-trade strategy.

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